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|-------------------|--|----|-----|
| 12 (a) (i) | 1. signal has same variation (with time) as the data | B1 | |
| | 2. consists of (a series of) 'highs' and 'lows' | B1 | |
| | <i>either</i> analogue is continuously variable (between limits) | | |
| | <i>or</i> digital has no intermediate values | B1 | [3] |
| (ii) | e.g. can be regenerated / noise can be eliminated | | |
| | extra data can be added to check / correct transmitted signal | | |
| | (any two reasonable suggestions, 1 each) | B2 | [2] |
| (b) (i) | analogue signal is sampled at (regular time) intervals | B1 | |
| | sampled signal is converted into a binary number | B1 | [2] |
| (ii) | one channel is required for each bit (of the digital number) | B1 | [1] |